

ZACHARY T. MINEIKO

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EDUCATION

University of Iowa

Bachelor of Science in Engineering - Electrical Engineering

May 2025

Focus Area: Computer Science

Related Coursework: Embedded Systems, Power Systems, Communication Networks, Computer Architecture

TECHNICAL SKILLS

Circuit Design & Simulation: Autodesk Fusion 360, AutoCAD, KiCad, Micro-Cap SPICE, PowerWorld, Powersim

Development Tools: AVR Assembly, SolidWorks, Atmel Studio, Matplotlib, Arduino IDE, Platform IO

Microcontrollers: ESP32, ATmega328p, ATtiny45, Arduino (Mega, Nano, Uno), Raspberry Pi

OS: MacOS, Linux (Kali Linux, Ubuntu, Debian), Windows

WORK EXPERIENCE

Outsider - Full Stack Developer

Dec 2024 – June 2025

- Architected and deployed novel cross-platform web application integrating React Native with Expo, and Supabase
- Led 3-person development team using sprint based workflows and task delegation through weekly planning meetings
- Established standardized development workflows to improve team consistency and code quality resulting in fewer integration conflicts and a smoother sprint cycle

Firefly - Engineering Technician Intern

Chicago, IL

Mar 2022 – Aug 2022

- Gained expertise in firmware and hardware troubleshooting for digital billboards on taxis and rideshare cars
- Transformed Chicago branch performance from worst to best performing through systematic repair methodologies
- Automated data collection processes using Python and OpenCV for gaining key insights on the taxi market
- Engineered repair workflows resulting in 20% increase in daily fixed screens stressing diligence and organization
- Participated in Agile processes by contributing to weekly stand-ups and observing development sprints

Custom Craft Builders - Electrician's Assistant/Web Developer

Chicago, IL

May 2020 – May 2023

- Lead installation teams for low voltage cable systems including coaxial and Ethernet network infrastructure
- Developed, deployed, and maintained company website using HTML, CSS, JavaScript, and Bootstrap 5 framework
- Configured and managed AWS EC2 instance hosting company website, ensuring high availability and performance
- Implemented automated messaging features on the website, improving client engagement and communication

PROJECT HIGHLIGHTS

Smart Pet Feeder - Senior Design Project

Dec 2024 – May 2025

- An IoT smart food tracking system for multi-pet homes to monitor and improve pets' health based on various metrics
- Developed multi-voltage ESP32 based control system with real time data processing and wireless connectivity
- Created custom PCB schematics and component footprints using Autodesk Fusion 360 for multi-sensor integration
- Integrated RFID identification system for individual pet recognition and personalized feeding schedules

IR Receiver with Hybrid Signal Filtering

Oct 2024

- Engineered and implemented a hybrid analog/digital IR receiver detecting IR interruptions using 940nm photodiode
- Designed multi-stage signal conditioning with non-inverting amplification, buffering, and a 2nd-order bandpass filter centered at ~421Hz to match existing emitter frequency constraint
- Implemented a 6th-order Chebyshev digital filter to extract signal envelope and reject ambient noise
- Validated system performance in high noise environments and achieved consistent detection up to 42 inches

High Performance Electric Longboard Battery Pack

July 2022

- Designed and constructed custom lithium-ion battery pack using 20 Molicel P42A 21700 cells in 10s2p configuration
- Engineered 29% capacity increase from 6.5Ah to 8.4Ah while maintaining 36V system voltage compatibility
- Solved mechanical packaging constraints fitting 20 cells and BMS wiring within existing enclosure

EXTRACURRICULAR ACTIVITIES & LEADERSHIP

CommUnity Food Pantry Volunteer

Iowa City, IC

Apr 2024 – June 2024

- Prepared and stocked food shelves, organized and monitored food tickets, trash management

Marist Robotics - Software Captain

Chicago, IL

Jan 2019

- Orchestrated development of software for Vex Robotics competitions competing against Chicagoland area schools

Illinois Institute of Technology – Code camp.

Chicago, IL

June – Aug 2018

- Led team developing Ninja parkour game displaying unique mechanics and physics using Unity and C#
- Runner up in one week game development competition